

INTERNAL

Growth and inflation pulse from ECB and Eurostat releases tested against SPY forward returns.

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N 10		EDGE +2.98%		HIT-RATE 80.0%	
CLASSIFICATION internal				AUTHOR Cerberus Research	
RUN ID 20260807_094827				GIT COMMIT 978cf61e	
DURATION 81.5s				DATASETS ecb/ecb_curated, eurostat/eurostat_curated	
PARAMETER · LOOKBACK_YEARS 15				PARAMETER · HORIZON 21	
KPI Value				N 10	
EDGE +2.98%				HIT-RATE 80.0%	
KEY FINDINGS · Latest Eurozone macro pulse: 219.426 · Latest growth: 8.434 · Latest inflation: 62.194					

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Data Sources

- **ecb/ecb_curated**: ECB monetary and real economy series
- **eurostat/eurostat_curated**: Eurostat activity and labour market series

Key Findings

- Latest Eurozone macro pulse: 219.426
- Latest growth: 8.434
- Latest inflation: 62.194
- Latest liquidity: 219.426

Executive Summary

NOTE

ECB / Eurozone macro pulse study. Combines growth, inflation, and liquidity indicators from ECB and Eurostat into a composite pulse, then tests SPY forward returns by pulse quintile.

Results & Interpretation

- Across 10 events, the forward return averaged +2.98%. Returns were positive 80.0% of the time.
- Across 14 events, the forward return averaged -0.69%. Returns were positive 57.1% of the time.
- Across 12 events, the forward return averaged +0.45%. Returns were positive 66.7% of the time.
- Across 6 events, the forward return averaged +1.89%. Returns were positive 83.3% of the time.
- Across 5 events, the forward return averaged +4.52%. Returns were positive 80.0% of the time.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Growth and inflation series are converted to YoY changes; liquidity series are normalized. The pulse is the cross-sectional average, then z-scored vs trailing 36-month history.

Charts

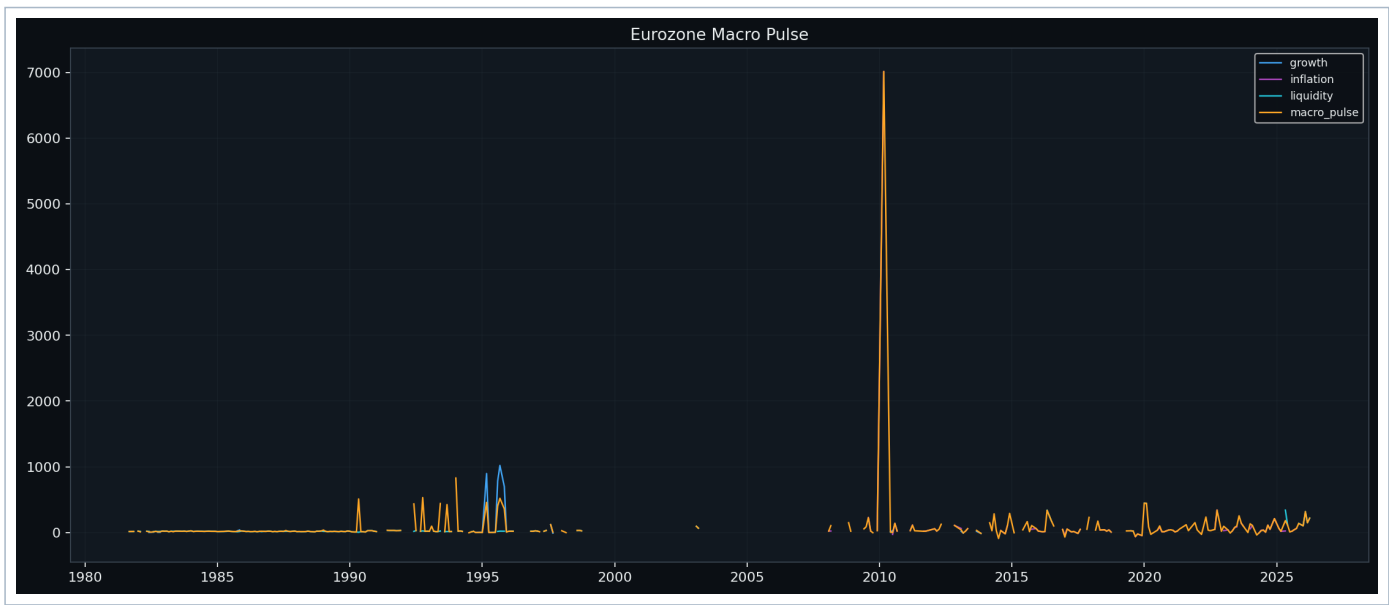


Figure 1. Eurozone macro pulse
Eurozone macro pulse

Statistical Results

macro_pulse	pulse_z
23.44	-0.26
28.31	-0.25
28.79	-0.25
112.27	-0.19
26.98	-0.24
19.51	-0.24
20.96	-0.24
57.80	-0.21
22.26	-0.23
49.72	-0.20
125.99	-0.15
108.05	-0.17
39.90	-0.21
-7.29	-0.24
20.82	-0.22
58.73	-0.19
32.61	-0.20

macro_pulse	pulse_z
-13.03	-0.24
148.58	-0.11
27.04	-0.20
284.41	-0.01
27.79	-0.20
-90.23	-0.29
31.02	-0.20
-19.60	-0.23
105.83	-0.14
291.45	2.90
-3.01	-0.74
40.85	-0.18
166.89	1.29
13.20	-0.56
106.48	0.55
61.17	0.00
22.45	-0.48
16.78	-0.51
11.12	-0.57
15.16	-0.52
338.20	2.85
172.88	1.05
99.94	0.27
47.03	-0.25
-69.09	-1.37
53.85	-0.13
9.42	-0.56
16.62	-0.47
-13.33	-0.78
48.98	-0.15
50.32	-0.13
230.86	1.58

macro_pulse	pulse_z
37.52	-0.33
174.56	1.00
38.56	-0.33
42.52	-0.24
21.72	-0.46
40.00	-0.32
7.01	-0.67
24.81	-0.48
26.44	-0.41
23.51	-0.46
-61.90	-1.44
-21.43	-0.92
-45.83	-1.14
446.83	3.56
444.41	2.92
87.24	0.11
-26.95	-0.77
36.42	-0.28
100.84	0.20
12.16	-0.49
12.01	-0.44
37.53	-0.20
42.34	-0.14
33.19	-0.21
9.30	-0.41
22.39	-0.33
52.66	-0.06
118.23	0.49
28.43	-0.31
151.22	0.73
39.21	-0.25
-28.44	-0.82

macro_pulse	pulse_z
235.72	1.41
35.91	-0.25
28.55	-0.31
47.63	-0.15
341.67	2.27
19.25	-0.41
95.27	0.21
39.43	-0.26
-7.52	-0.63
26.12	-0.36
76.99	0.06
100.71	0.25
254.67	1.45
140.79	0.49
93.62	0.07
0.09	-0.72
133.27	0.36
106.67	0.25
-37.22	-1.31
-7.14	-0.89
28.67	-0.48
37.55	-0.37
5.36	-0.73
112.69	0.56
49.18	-0.24
211.96	1.66
62.06	-0.14
18.03	-0.66
182.37	1.23
110.92	0.36
8.43	-0.83
20.87	-0.65

macro_pulse	pulse_z
44.08	-0.39
62.19	-0.15
139.55	0.73
102.26	0.26
318.43	2.67
148.65	0.70
219.43	1.41

Table 1. Eurozone macro pulse

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	10	+ 2.98%	+ 3.35%	+ 80.00%
Q2	14	-0.69%	+ 0.58%	+ 57.14%
Q3	12	+ 0.45%	+ 1.85%	+ 66.67%
Q4	6	+ 1.89%	+ 3.71%	+ 83.33%
Q5	5	+ 4.52%	+ 5.43%	+ 80.00%

Table 2. SPY forward returns by pulse quintile

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

pulse

timestamp	macro_pulse	pulse_z
2010-09-01	23.44	-0.26
2010-12-01	28.31	-0.25
2011-03-01	28.79	-0.25
2011-04-01	112.27	-0.19
2011-05-01	26.98	-0.24
2011-09-01	19.51	-0.24
2011-10-01	20.96	-0.24
2012-02-01	57.80	-0.21
2012-03-01	22.26	-0.23

timestamp	macro_pulse	pulse_z
2012-04-01	49.72	-0.20
2012-05-01	125.99	-0.15
2012-11-01	108.05	-0.17
2013-02-01	39.90	-0.21
2013-03-01	-7.29	-0.24
2013-04-01	20.82	-0.22

spy_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	10	+ 2.98%	+ 3.35%	+ 80.00%
Q2	14	-0.69%	+ 0.58%	+ 57.14%
Q3	12	+ 0.45%	+ 1.85%	+ 66.67%
Q4	6	+ 1.89%	+ 3.71%	+ 83.33%
Q5	5	+ 4.52%	+ 5.43%	+ 80.00%

Statistics Summary

Key	Value
study	ecb_eurozone_macro_pulse
category	macro
status	success
parameters	{'lookback_years': 15, 'horizon': 21}
datasets	['ecb/ecb_curated', 'eurostat/eurostat_curated']
rows	10865
git_commit	978cf61eefd9b125a5087073e70ef1c9d91f687b
executed_at	2026-08-07T08:48:27.812469+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.

Term	Definition
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-07 08:48 UTC
Author	Cerberus Research
Study slug	ecb_eurozone_macro_pulse
Run ID	20260807_094827
Git commit	978cf61e
Datasets	ecb/ecb_curated, eurostat/eurostat_curated
Parameters hash	0cdb4e94b3cb
Parameter · lookback_years	15
Parameter · horizon	21
Data fingerprint	fa371ec3ec9f
Duration	81.5s